Chapter 4

Initial-State Geometry and the Ultracentral Flow Puzzle in ^{208}Pb+^{208}Pb Collisions

This chapter describes the computational framework developed to investigate the ultracentral ^{208}Pb+^{208}Pb flow puzzle through a systematic scan of the nuclear initial-state parameter space. We introduce the theoretical foundations of initial-state geometry, the nuclear structure model, the TRENTO *entropy-deposition framework, and the statistical design methodology. The full pipeline — from Latin Hypercube Sampling through eccentricity computation — is described in both mathematical and algorithmic terms.*

4.1 Introduction and Motivation

The collective flow of matter produced in ultrarelativistic heavy-ion collisions at the Large Hadron Collider (LHC) is among the most precisely studied emergent phenomena in nuclear physics. In non-central collisions ($b \gtrsim 2\,\mathrm{fm}$), the initial almond-shaped overlap region of the two colliding nuclei generates a dominant elliptic anisotropy in momentum space, characterised by $v_2\{2\}$. As the collision centrality approaches zero, the geometric contribution of the impact parameter vanishes, and all flow harmonics are driven solely by quantum fluctuations in nucleon positions within the individual nuclei.

In this regime — hereafter referred to as the *ultracentral* class, operationally defined as the 0–1% most central collisions — a cluster of persistent discrepancies between experimental data and state-of-the-art hydrodynamic model calculations has been identified. Collectively, these discrepancies constitute the **ultracentral flow puzzle**:

The Ultracentral Flow Puzzle — Experimental Signatures

(i) **v_2-to-v_3 hierarchy reversal.** Viscous hydrodynamic calculations universally predict $v_2\{2\} > v_3\{2\}$ in the 0–1% centrality bin, because the elliptic eccentricity ε_2 retains a small finite mean from the residual geometry. ALICE and ATLAS measurements instead show $v_2\{2\} \approx v_3\{2\}$, implying an unexpected suppression of elliptic flow or enhancement of triangular flow [1, 2, 3].

(ii) **Negative $v_4\{4\}^4$.** The four-particle hexadecapole cumulant combination $v_4\{4\}^4 \equiv 2\langle v_4^2\rangle^2 - \langle v_4^4\rangle$ is measured to be *negative* in ^{208}Pb+^{208}Pb at $\sqrt{s_{\mathrm{NN}}} = 2.76$ and $5.02\,\mathrm{TeV}$ by ALICE, ATLAS, and CMS [4, 5, 6]. Standard models predict a positive value.

(iii) **Small $v_2\{4\}/v_2\{2\}$ ratio.** The fluctuation-suppression ratio $R_{42} \equiv v_2\{4\}/v_2\{2\}$ approaches values below 0.5 in the 0–0.1% bin, significantly smaller than model predictions, indicating a non-Gaussian distribution $p(\varepsilon_2)$ with much lighter tails than expected [7].

(iv) **Enhanced $v_3\{4\}/v_3\{2\}$ ratio.** Models systematically underpredict this ratio in ultra-central collisions, i.e. they overestimate the relative width of $p(\varepsilon_3)$ [8].

Proposed resolutions involve modifications to the nuclear structure of ^{208}Pb: a small octupole deformation β_3 [9], a hexadecapole component β_4 coupled to the nonlinear mode-mixing channel $v_4 \sim \chi_{42}\, v_2^2$ [10], short-range nucleon-nucleon correlations [11], and modifications to the nucleon substructure (Trento parameters w, k). The primary goal of the present study is to perform a systematic, quantitative scan of this parameter space using a Latin Hypercube design, enabling subsequent Gaussian Process emulation and Bayesian inference.

4.2 Nuclear Structure: The Deformed Woods-Saxon Model

4.2.1 The Woods-Saxon Density Profile

The nuclear one-body density of a spherically symmetric nucleus of mass number A is well described by the Fermi distribution, commonly known as the Woods-Saxon (WS)

profile [12]:

$$\rho_0(r) = \frac{\rho_{\text{sat}}}{1 + \exp\left(\dfrac{r - R}{a}\right)}, \tag{4.1}$$

where $\rho_{\text{sat}} \approx 0.16\,\text{fm}^{-3}$ is the nuclear saturation density, R is the half-density radius, and a is the diffuseness parameter governing the surface-to-interior transition. For ^{208}Pb the empirical values extracted from electron-scattering experiments are $R = 6.624\,\text{fm}$ and $a = 0.549\,\text{fm}$ [13].

4.2.2 Deformed Density Profiles

For a deformed nucleus, the surface is described by a multipole expansion in real spherical harmonics $Y_\ell^0(\theta, \phi)$. The deformed WS radius is

$$R(\theta, \phi) = R\left[1 + \sum_{\ell=2}^{\infty} \beta_\ell\, Y_\ell^0(\theta, \phi)\right], \tag{4.2}$$

where β_ℓ is the dimensionless deformation parameter of multipolarity ℓ, and the sum runs over even and odd multipoles as permitted by nuclear symmetry. Substituting eq. (4.2) into eq. (4.1) gives the deformed density:

$$\rho(\boldsymbol{r}) = \frac{\rho_{\text{sat}}}{1 + \exp\left(\dfrac{|\boldsymbol{r}| - R(\theta, \phi)}{a}\right)}. \tag{4.3}$$

The deformation parameters relevant to this study are:

- **Quadrupole β_2:** The leading-order prolate/oblate deformation. For doubly-magic ^{208}Pb, the ground-state quadrupole moment is negligible: $\beta_2 \approx 0$ [15].
- **Octupole β_3:** A pear-shaped deformation that breaks reflection symmetry. Although ^{208}Pb is doubly magic, low-energy nuclear structure calculations permit a small dynamic octupole component. Data from ALICE and ATLAS constrain $\beta_3 \lesssim 0.0375$ for static deformation [9]; recent dynamical proposals allow an effective value up to $\beta_3 \sim 0.07$ [17].
- **Hexadecapole β_4:** Couples to the fourth-order flow coefficient via the nonlinear channel. DFT calculations give $|\beta_4| \sim 0.01$–0.02 for ^{208}Pb [16].

For a nucleus undergoing a random three-dimensional rotation before each collision (as appropriate for a quantum ground state), the deformation enters the initial-state geometry only through the event-by-event orientation average. In the isobar-sampler framework [25], an Euler rotation is applied to each generated nuclear configuration.

4.2.3 Short-Range Nucleon-Nucleon Correlations

Beyond the mean-field WS profile, nucleons interact via a hard-core short-range repulsion at distances $r \lesssim 0.4\,\mathrm{fm}$, arising from Pauli blocking and the repulsive core of the nucleon-nucleon potential. This suppresses nucleon clustering at short distances and modifies the sub-nucleonic granularity of the initial state.

The correlation function is implemented in the isobar sampler as a step function:

$$C(r_{ij}) = \begin{cases} C_s & \text{if } r_{ij} < C_\ell, \\ 0 & \text{if } r_{ij} \geq C_\ell, \end{cases} \tag{4.4}$$

where r_{ij} is the separation between nucleons i and j, C_ℓ is the correlation length (hard-core radius), and $C_s \in [-1, 0]$ is the correlation strength. The value $C_s = -1$ corresponds to full Pauli exclusion (no two nucleons within C_ℓ of each other); $C_s = 0$ is the uncorrelated Woods-Saxon case. Following [25], we fix $C_\ell = 0.4\,\mathrm{fm}$ and $C_s = -1$ as the default values.

4.2.4 The Isobar Sampler

Generating nuclear configurations with varied WS parameters independently for each design point would introduce uncorrelated statistical noise that obscures the genuine physical response to parameter changes. The isobar-sampler framework of Luzum et al. [25] addresses this by factorising the stochastic content into a *seed bank*.

Stage 1: Seed generation. A bank of N_{conf} *seeds* — vectors of independent random numbers in $[0, 1]$ — is generated once and stored. These seeds encode the relative positions of nucleons in a parameter-independent space. Formally, nucleon position $\boldsymbol{r}_i$ is computed as a deterministic function of the WS parameters and the corresponding seed ξ_i:

$$\boldsymbol{r}_i = \mathcal{F}(R, a, \beta_2, \beta_3, \beta_4, C_\ell, C_s;\ \xi_i)\,. \tag{4.5}$$

Stage 2: Configuration generation. For each design point $\theta = (R, a, \beta_3, \beta_4, \ldots)$, the same seed bank is used with the new parameters to produce N_{conf} nuclear configurations. Because the mapping eq. (4.5) is monotone in each seed component, configurations generated from the same seed bank with different parameters differ only through their physical response and not through spurious statistical fluctuations. The N_{conf} configurations are written to an HDF5 file (e.g. `WS1.hdf`) for use by TRENTo.

4.3 Initial-State Entropy Deposition: The TRENTo Model

4.3.1 General Entropy-Deposition Ansatz

The TRENTo (Reduced Thickness Event-by-event Nuclear Topology) model [20] provides a flexible parametric ansatz for the initial-state entropy density $s(\boldsymbol{x}_\perp)$ in the transverse plane. The fundamental object is the *reduced thickness function*:

$$T_R(p;\, T_A, T_B) = \left(\frac{T_A^p + T_B^p}{2}\right)^{1/p}, \tag{4.6}$$

where $T_A(\boldsymbol{x}_\perp)$ and $T_B(\boldsymbol{x}_\perp)$ are the participant thickness functions of the two nuclei evaluated at transverse position $\boldsymbol{x}_\perp$, and p is the entropy-deposition exponent. The entropy density is then

$$s(\boldsymbol{x}_\perp) = \kappa\, T_R(p;\, T_A, T_B), \tag{4.7}$$

with κ an overall normalisation constant.

The parameter p interpolates between well-known limiting cases:

$$p \to -\infty: \quad T_R \propto \min(T_A, T_B) \quad \text{(wounded nucleon, target)}, \tag{4.8}$$

$$p = -1: \quad T_R \propto \frac{2\,T_A T_B}{T_A + T_B} \quad \text{(harmonic mean)}, \tag{4.9}$$

$$p = 0: \quad T_R \propto \sqrt{T_A T_B} \quad \text{(geometric mean, KLN-like)}, \tag{4.10}$$

$$p = +1: \quad T_R \propto \frac{T_A + T_B}{2} \quad \text{(arithmetic mean, WN-like)}. \tag{4.11}$$

Bayesian analyses of LHC Pb+Pb data prefer values near $p \approx 0$ [21, 22].

4.3.2 Nucleon Sub-Structure

The participant thickness function of nucleus A is the sum over participant nucleon contributions:

$$T_A(\boldsymbol{x}_\perp) = \sum_{i \in \text{part.}} \gamma_i\, \rho_{\text{nuc}}(\boldsymbol{x}_\perp - \boldsymbol{s}_i), \tag{4.12}$$

where $\boldsymbol{s}_i$ is the transverse position of nucleon i, γ_i is an independent gamma-distributed fluctuation,

$$\gamma_i \sim \text{Gamma}(k, k^{-1}), \quad \langle \gamma_i \rangle = 1, \quad \text{Var}(\gamma_i) = k^{-1}, \tag{4.13}$$

and ρ_{nuc} is the nucleon density profile. Each nucleon is modelled as a Gaussian:

$$\rho_{\text{nuc}}(\boldsymbol{x}_\perp) = \frac{1}{2\pi w^2} \exp\left(-\frac{|\boldsymbol{x}_\perp|^2}{2w^2}\right), \tag{4.14}$$

where w is the nucleon width in the transverse plane. Smaller w produces a grainier, more fluctuation-dominated initial state.

Two nucleons i (in nucleus A) and j (in nucleus B) are considered to have interacted (participated) if their transverse separation satisfies

$$|\boldsymbol{s}_i - \boldsymbol{s}_j|^2 < d^2, \tag{4.15}$$

where d is the minimum nucleon distance parameter.

4.3.3 Fixed Trento Parameters

The Bayesian analyses of [21] and [22] provide well-constrained posterior distributions for the Trento parameters in Pb+Pb. In this study we fix:

$$p = 0.0, \quad k = 1.6, \quad d = 1.0\,\mathrm{fm}, \quad \mathrm{norm} = \begin{cases} 18.0 & \sqrt{s_{\mathrm{NN}}} = 2.76\,\mathrm{TeV}, \\ 20.0 & \sqrt{s_{\mathrm{NN}}} = 5.02\,\mathrm{TeV}. \end{cases} \tag{4.16}$$

The energy-dependent normalisation reflects the approximately $11\,\%$ increase in charged-particle pseudorapidity density $dN_{\mathrm{ch}}/d\eta$ between the two LHC energies [23, 24]. Only the nucleon width w is kept free, as it directly controls the sub-nucleonic granularity of the initial state and hence the fluctuation spectrum of the eccentricities.

4.4 Eccentricity Cumulants and the Flow Observables

4.4.1 Spatial Eccentricity Vector

The spatial anisotropy of the initial-state entropy density is characterised by the complex eccentricity vector of order n:

$$\mathcal{E}_n = \varepsilon_n\, e^{in\Psi_n} \equiv -\frac{\langle r^n\, e^{in\phi}\rangle_s}{\langle r^n\rangle_s}, \tag{4.17}$$

where the bracket $\langle\cdot\rangle_s$ denotes the entropy-weighted two-dimensional average over the transverse plane:

$$\langle f\rangle_s = \frac{\int f(\boldsymbol{x}_\perp)\, s(\boldsymbol{x}_\perp)\, d^2\boldsymbol{x}_\perp}{\int s(\boldsymbol{x}_\perp)\, d^2\boldsymbol{x}_\perp}, \tag{4.18}$$

(r, ϕ) are polar coordinates in the participant-centred frame, $\varepsilon_n = |\mathcal{E}_n| \in [0, 1]$ is the eccentricity magnitude, and Ψ_n is the n-th order participant-plane angle. The r^n weighting (Teaney–Yan prescription [18]) ensures a linear response between ε_n and v_n for $n = 2, 3$ at leading order in viscous corrections.

4.4.2 Multi-Particle Cumulants of the Eccentricity Distribution

Let ε_n denote the magnitude of the eccentricity in a given event. Across an ensemble of events within a centrality class, the distribution $p(\varepsilon_n)$ encodes both the mean geometry and its fluctuations. Multi-particle flow cumulants measure moments of this distribution in a way that is robust against non-flow correlations [19]. The eccentricity cumulants are defined through the moment generating function:

$$\varepsilon_n\{2\} = \sqrt{\langle \varepsilon_n^2 \rangle}, \tag{4.19}$$

$$\varepsilon_n\{4\} = \left(2\langle \varepsilon_n^2 \rangle^2 - \langle \varepsilon_n^4 \rangle\right)^{1/4}, \tag{4.20}$$

$$\varepsilon_n\{6\} = \left(-\frac{c_6}{4}\right)^{1/6}, \quad c_6 = \langle \varepsilon_n^6 \rangle - 9\langle \varepsilon_n^4 \rangle \langle \varepsilon_n^2 \rangle + 12\langle \varepsilon_n^2 \rangle^3, \tag{4.21}$$

$$\varepsilon_n\{8\} = \left(\frac{c_8}{33}\right)^{1/8}, \quad c_8 = \langle \varepsilon_n^8 \rangle - 16\langle \varepsilon_n^6 \rangle \langle \varepsilon_n^2 \rangle - 18\langle \varepsilon_n^4 \rangle^2 + 144\langle \varepsilon_n^4 \rangle \langle \varepsilon_n^2 \rangle^2 - 144\langle \varepsilon_n^2 \rangle^4. \tag{4.22}$$

The convergence $\varepsilon_n\{4\} = \varepsilon_n\{6\} = \varepsilon_n\{8\}$ signals a Gaussian distribution $p(\varepsilon_n)$. Deviations indicate non-Gaussianity, which is the hallmark of the ultracentral regime.

4.4.3 Puzzle-Critical Observables

We now define the four observables that directly diagnose the ultracentral flow puzzle, together with their eccentricity proxies.

Triangularity Ratio T_{32}

The ratio

$$T_{32} \equiv \frac{\varepsilon_3\{2\}}{\varepsilon_2\{2\}} = \frac{\sqrt{\langle \varepsilon_3^2 \rangle}}{\sqrt{\langle \varepsilon_2^2 \rangle}} \tag{4.23}$$

tracks the transition from the geometry-dominated regime ($T_{32} \ll 1$ in peripheral collisions, where ε_2 is large) to the fluctuation-dominated ultracentral regime. Spherical models predict $T_{32} \lesssim 0.8$ in the 0–1% bin; ALICE data are consistent with $T_{32} \approx 1.0$ (see Sec. 4.1).

Fluctuation-Suppression Ratio R_{42}

$$R_{42} \equiv \frac{\varepsilon_2\{4\}}{\varepsilon_2\{2\}} = \left(\frac{2\langle \varepsilon_2^2 \rangle^2 - \langle \varepsilon_2^4 \rangle}{\langle \varepsilon_2^2 \rangle^2}\right)^{1/4} \tag{4.24}$$

is bounded $R_{42} \leq 1$ and equals unity in the absence of fluctuations. For a distribution with mean $\bar{\varepsilon}_2$ and variance σ^2:

$$R_{42} \approx \frac{\bar{\varepsilon}_2}{\sqrt{\bar{\varepsilon}_2^2 + \sigma^2}}. \tag{4.25}$$

As $\bar{\varepsilon}_2 \to 0$ in ultra-central collisions, $R_{42} \to 0$ unless σ^2 also vanishes. Data in the 0–0.1% bin show $R_{42} < 0.5$, indicating a strongly non-Gaussian $p(\varepsilon_2)$ with lighter tails than any current model predicts.

Normalised Kurtosis Γ_2

The normalised excess kurtosis of $p(\varepsilon_2)$,

$$\Gamma_2 \equiv \frac{\langle \varepsilon_2^4 \rangle - 2\langle \varepsilon_2^2 \rangle^2}{\langle \varepsilon_2^2 \rangle^2}, \tag{4.26}$$

satisfies $\Gamma_2 = 0$ for a Gaussian, $\Gamma_2 > 0$ for heavier tails (leptokurtic), and $\Gamma_2 < 0$ for lighter tails (platykurtic). In the ultracentral limit where $p(\varepsilon_2)$ approaches a Rayleigh distribution, $\Gamma_2 \to -2$, i.e. $\langle \varepsilon_2^4 \rangle \to 2\langle \varepsilon_2^2 \rangle^2/(1+1) = \langle \varepsilon_2^2 \rangle^2$. The relation to the fluctuation-suppression ratio is:

$$R_{42}^4 = 2 + \Gamma_2. \tag{4.27}$$

Hexadecapole Sign-Change Observable $\varepsilon_4\{4\}^4$

The four-particle hexadecapole cumulant combination:

$$\varepsilon_4\{4\}^4 \equiv 2\langle \varepsilon_4^2 \rangle^2 - \langle \varepsilon_4^4 \rangle \tag{4.28}$$

is the direct eccentricity analogue of the measured $v_4\{4\}^4$. This quantity becomes *negative* when the distribution $p(\varepsilon_4)$ is sufficiently broad, i.e. when $\langle \varepsilon_4^4 \rangle > 2\langle \varepsilon_4^2 \rangle^2$.

The mechanism driving $v_4\{4\}^4 < 0$ involves the nonlinear mode-coupling channel $v_4 \approx \chi_{42}\, v_2^2$, where χ_{42} is the linear response coefficient. The contribution of this channel to $v_4\{4\}^4$ is negative whenever [10]:

$$\frac{\langle \varepsilon_2^8 \rangle}{\langle \varepsilon_2^4 \rangle^2} > 2. \tag{4.29}$$

This ratio — denoted `nl_r8` in the code — therefore serves as a *sign-change trigger*: if it exceeds 2, the predicted $v_4\{4\}^4$ will be negative even for a model with vanishing

mean ε_4. The nonlinear eccentricity coupling is:

$$\varepsilon_4^{\Psi_2} \equiv \frac{\langle \varepsilon_4 \, \varepsilon_2^2 \rangle}{\sqrt{\langle \varepsilon_2^4 \rangle}}, \tag{4.30}$$

which measures the correlation between the hexadecapole and elliptic eccentricities in the same event.

4.4.4 Summary of Computed Observables

Table 4.1 summarises the complete set of eccentricity observables computed for each design point in the 0–1% ultracentral bin.

Table 4.1: Eccentricity observables computed per design point in the 0–1% centrality bin. All are evaluated using the event ensemble from a single Trento run. "KEY" denotes a puzzle-critical quantity.

Code key	Mathematical definition	Physical meaning	Puzzle?
`e2_2`	$\sqrt{\langle \varepsilon_2^2 \rangle}$	Elliptic eccentricity (RMS)	
`e2_4`	$(2\langle \varepsilon_2^2 \rangle^2 - \langle \varepsilon_2^4 \rangle)^{1/4}$	4-particle cumulant	
`e2_6`	$(-c_6/4)^{1/6}$	6-particle cumulant	
`e2_8`	$(c_8/33)^{1/8}$	8-particle cumulant	
`e3_2`	$\sqrt{\langle \varepsilon_3^2 \rangle}$	Triangularity (RMS)	KEY
`e4_2`	$\sqrt{\langle \varepsilon_4^2 \rangle}$	Hexadecapole (RMS)	
`R42`	$\varepsilon_2\{4\}/\varepsilon_2\{2\}$	Fluctuation suppression	KEY
`T32`	$\varepsilon_3\{2\}/\varepsilon_2\{2\}$	Triangularity ratio	KEY
`Gamma2`	$(\langle \varepsilon_2^4 \rangle - 2\langle \varepsilon_2^2 \rangle^2)/\langle \varepsilon_2^2 \rangle^2$	Excess kurtosis	KEY
`e4_4c`	$2\langle \varepsilon_4^2 \rangle^2 - \langle \varepsilon_4^4 \rangle$	Hexadecapole sign-change	KEY
`nl_r8`	$\langle \varepsilon_2^8 \rangle / \langle \varepsilon_2^4 \rangle^2$	NL sign-change trigger	KEY

4.5 Centrality Determination

Centrality is determined from the charged-particle multiplicity distribution $dN/d\eta$ via a mapping from the Glauber model. In the TRENTO simulation, each event produces a multiplicity proxy proportional to $\int s(\boldsymbol{x}_\perp)\, d^2\boldsymbol{x}_\perp$. Events are ranked in descending order of multiplicity M, and the centrality percentile of event i is assigned as:

$$c_i = \frac{\mathrm{rank}(-M_i)}{N_{\mathrm{ev}}} \times 100\%, \tag{4.31}$$

where $\mathrm{rank}(-M_i)$ is the rank of event i in the descending multiplicity ordering and N_{ev} is the total number of events. The ultracentral bin is defined as $c \in [0\%, 1\%]$, and

the bin population for $N_{\rm ev} = 5 \times 10^5$ is $N_{\rm UC} \approx 5000$ events, sufficient for a statistical uncertainty of $\lesssim 1\%$ on all cumulant observables after bootstrap resampling.

4.6 Statistical Design: The Latin Hypercube

4.6.1 The Design Problem

We wish to evaluate the map

$$\boldsymbol{\theta} = (R,\, a,\, \beta_3,\, \beta_4,\, w) \longrightarrow \boldsymbol{y}(\boldsymbol{\theta}) = \big(\varepsilon_2\{2\},\, T_{32},\, R_{42},\, \Gamma_2,\, \varepsilon_4\{4\}^4, \ldots\big) \tag{4.32}$$

at $N_{\rm d}$ design points $\{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(N_{\rm d})}\}$ distributed across the five-dimensional parameter space:

$$\boldsymbol{\theta} \in \Theta \equiv [6.50, 6.80] \times [0.44, 0.65] \times [0.00, 0.12] \times [-0.02, 0.06] \times [0.40, 1.20] \quad \text{(fm, fm, —, —, fm)}. \tag{4.33}$$

These ranges are motivated by the nuclear data compilations of [13, 9, 16] and the Trento Bayesian analysis of [21].

4.6.2 Latin Hypercube Sampling

Definition 4.6.1 (Latin Hypercube Sample)**.** *A* Latin Hypercube Sample *(LHS) of size N in d dimensions is a matrix $\mathbf{U} \in [0,1]^{N \times d}$ such that, for each column $j = 1, \ldots, d$, the entries $\{U_{1j}, \ldots, U_{Nj}\}$ form a* stratified *sample: each stratum $[(k-1)/N,\, k/N)$ for $k = 1, \ldots, N$ contains exactly one entry.*

A LHS achieves uniformity in every one-dimensional projection, which is necessary but not sufficient for good coverage of the full d-dimensional space. To improve space-filling, we employ the *maximin* criterion [26]:

$$\mathbf{U}^* = \underset{\mathbf{U} \in \mathcal{L}_N^d}{\operatorname{argmax}} \min_{i \neq j} \|\boldsymbol{u}_i - \boldsymbol{u}_j\|_2, \tag{4.34}$$

where $\mathcal{L}_N^d$ is the set of all N-point LHS designs in d dimensions, and $\boldsymbol{u}_i$ is the i-th row of $\mathbf{U}$.

Because an exact solution of eq. (4.34) is NP-hard for large N and d, we use a random search approximation: $N_{\rm iter} = 2000$ independent random LHS candidates are generated and the one with the largest minimum pairwise distance is retained.

Design matrix. The physical design is obtained by affine scaling:

$$\theta_j^{(i)} = \theta_j^{\min} + U_{ij}\left(\theta_j^{\max} - \theta_j^{\min}\right), \quad i = 1, \ldots, N;\; j = 1, \ldots, d. \tag{4.35}$$

4.6.3 Number of Design Points

For a subsequent Gaussian Process (GP) emulator of the map eq. (4.32) in $d = 5$ dimensions, the standard recommendation is $N_\mathrm{d} \geq 10\,d = 50$ design points as an absolute minimum, with $N_\mathrm{d} = 100$ providing adequate coverage for a squared-exponential kernel with $\mathcal{O}(d)$ hyper-parameters [27]. We choose $N_\mathrm{d} = 100$ as our baseline and note that $N_\mathrm{d} = 200$ would be preferable if the computational budget allows, as it better resolves non-linear structures in parameter space.

Remark 4.6.1. *The computational cost of the full pipeline scales as $N_\mathrm{d} \times t_\mathrm{Trento}$, where $t_\mathrm{Trento} \approx 15$–$30\,\mathrm{min}$ for 5×10^5 events on 32 CPU cores. For $N_\mathrm{d} = 100$, the total wall time is approximately 25–$50\,\mathrm{hours}$; this is the primary computational bottleneck.*

4.6.4 Space-Filling Quality Assessment

The quality of the LHS is assessed by three complementary metrics:

(a) **Marginal uniformity:** Histogram of each parameter's marginal distribution should be approximately flat.

(b) **Minimum pairwise distance:** $d_\mathrm{min} = \min_{i \neq j} \|\boldsymbol{u}_i - \boldsymbol{u}_j\|_2$ in the unit hypercube; larger is better.

(c) **2-D projections:** The lower triangle of the corner plot (scatter of all $\binom{d}{2}$ parameter pairs) should show no obvious clustering or regular structure.

The corner plot produced by the pipeline (`corner_lhs.pdf`) is the primary visual diagnostic.

4.7 The Computational Pipeline

4.7.1 Pipeline Architecture

The full analysis is implemented in the Python script `pbpb_ultracentral_scan.py`, which executes five sequentially dependent stages summarised in table 4.2. Each stage writes its output to a persistent cache so that any individual stage can be re-run independently by setting the corresponding `RUN_STAGE_*` flag to `True`.

4.7.2 Stage 0: Latin Hypercube Generation

The implementation of the maximin LHS follows algorithm 1. For $N_\mathrm{d} = 100$, $d = 5$, and $N_\mathrm{iter} = 2000$, the algorithm completes in $\mathcal{O}(1)\,\mathrm{s}$ on a modern CPU.

Table 4.2: Pipeline stages, inputs, and outputs.

#	Stage	Input	Output
0	LHS design generation	Configuration flags	`lhs_design_matrix.npz`
1	Seed bank creation	Seeds config YAML	`seeds/nucleon-seeds.hdf`
2	Nuclear configuration	`nucleon-seeds.hdf`, isobar config YAML	`design_NNNN/WS1.hdf`, `design_NNNN/WS2.hdf`
3	TRENTO simulation	`WS*.hdf`, fixed Trento params	`design_NNNN/trento_events.npy`
4	Eccentricity computation	`trento_events.npy`	`results/observables.npz`
5	Visualisation	`observables.npz`	`results/plots/*.pdf`

Algorithm 1: Maximin-Optimised Latin Hypercube Sampling

Input: N (design points), d (parameters), seed, N_{iter} (trials)

Output: $\mathbf{U}^* \in [0,1]^{N\times d}$ (unit-cube design)

```
d_best ← −∞;
for t = 1 to N_iter do
    foreach column j = 1, ..., d do
        Draw π_j ← Permutation(N);
        U_ij ← (π_j[i] + u_ij)/N,;
                where u_ij ~ U[0,1]
    end
    d_min ← min_{i≠j} ||u_i − u_j||_2;
    if d_min > d_best then
        d_best ← d_min;;
        U* ← U;
    end
end
return U*
```

4.7.3 Stages 1–2: Nuclear Configuration Generation

For each design point $\boldsymbol{\theta}^{(i)}$, the following two-step procedure generates the nuclear HDF5 files for TRENTO:

Step 1 (once per campaign): The seeds configuration is written as:

```
isobar_seeds:
  number_nucleons:  {value: 208}
  number_configs:   {value: 10000}
  output_file:      {filename: nucleon-seeds.hdf}
  number_of_parallel_processes: {value: -1}
```

and `make_seeds.py` is invoked.

Step 2 (once per design point): The isobar configuration file maps the LHS sample to the isobar-sampler YAML schema. Both isobars (`WS1`, `WS2`) are assigned identical parameters, reflecting the symmetry of ^{208}Pb+^{208}Pb collisions:

```
isobar_properties:
  isobar1:
    isobar_name: WS1
    WS_radius:            {value: R[i]}
    WS_diffusiveness:     {value: a[i]}
    beta_2:               {value: 0.0}
    gamma:                {value: 0.0}
    beta_3:               {value: beta3[i]}
    beta_4:               {value: beta4[i]}
    correlation_length:   {value: 0.4}
    correlation_strength: {value: -1.0}
  isobar2:
    isobar_name: WS2
    [identical parameters]
```

4.7.4 Stage 3: TRENTo Simulation

TRENTo is called for each design point with the flag `--random-seed` $= 1000 + i$ to ensure independent event sequences across design points. The `-o` flag is deliberately *omitted* so that TRENTo writes only per-event summary lines to `stdout` rather than individual event files, eliminating N_{ev} costly disk writes. The `stdout` format is:

```
event_id   b   N_part   M   epsilon_2   epsilon_3   epsilon_4   epsilon_5
```

and the parsed array of shape $(N_{\mathrm{ev}}, 8)$ is cached as a compressed `.npy` file.

4.7.5 Stage 4: Eccentricity Computation and Bootstrapping

After centrality assignment via eq. (4.31), the N_{UC} ultracentral events are selected. All observables in table 4.1 are computed analytically from the sample moments $m_k \equiv N_{\mathrm{UC}}^{-1} \sum_{i=1}^{N_{\mathrm{UC}}} \varepsilon_2^k$.

Statistical uncertainties are estimated by the bootstrap [28]: $N_{\mathrm{boot}} = 300$ resamples with replacement are drawn from the N_{UC} events, observables are recomputed for each resample, and the 16th–84th percentile interval of the resulting bootstrap distribution is reported as the $\pm 1\sigma$ band. This correctly propagates the wider uncertainty in the small 0–0.1% bin relative to the 0–1% bin.

Algorithm 2: Bootstrap Uncertainty Estimation for Ultracentral Observables

Input: Events $\{e_1, \ldots, e_{N_{\mathrm{UC}}}\}$,;
N_{boot}
Output: Central values $\hat{y}$, lower $\hat{y}_-$, upper $\hat{y}_+$

```
ŷ ← ComputeObs({e_i});
for b = 1 to N_boot do
    I ← sample N_UC indices with replacement;
    ŷ^(b) ← ComputeObs({e_I});
end
ŷ_− ← Percentile_16({ŷ^(b)});
ŷ_+ ← Percentile_84({ŷ^(b)});
return (ŷ, ŷ_−, ŷ_+)
```

4.8 Parameter Space and Prior Ranges

Table 4.3 summarises the five free parameters, their prior ranges, physical motivation, and key references.

Table 4.3: Free parameters, prior ranges, central (reference) values, and literature references for the ^{208}Pb+^{208}Pb ultracentral scan.

Parameter	Symbol	Range	Central	Unit	Reference
WS radius	R	$[6.50,\ 6.80]$	6.624	fm	[13]
WS diffuseness	a	$[0.44,\ 0.65]$	0.549	fm	[13, 14]
Octupole	β_3	$[0.00,\ 0.12]$	0.0	—	[9, 17]
Hexadecapole	β_4	$[-0.02,\ 0.06]$	0.0	—	[16]
Nucleon width	w	$[0.40,\ 1.20]$	0.76	fm	[21]

The octupole range $\beta_3 \in [0, 0.12]$ is chosen to span the constraint of [9] ($\beta_3 \lesssim 0.0375$, which disfavours a static octupole as the sole resolution of the puzzle) and the more liberal range entertained by proposals involving a dynamical octupole "breathing mode" [17]. The upper end of the hexadecapole range $\beta_4 = 0.06$ exceeds current DFT predictions but is included to assess the sensitivity of $v_4\{4\}^4$ to large β_4 values.

4.9 Implementation Details and Code Structure

4.9.1 Directory Structure

The pipeline creates the following directory tree under `pbpb_scan_276GeV/` (or `...502GeV/` for $\sqrt{s_{\mathrm{NN}}} = 5.02\,\mathrm{TeV}$):

```
pbpb_scan_276GeV/
  lhs_design_matrix.npz          # design matrix + metadata
  seeds/
    seeds-conf.yaml               # isobar-sampler seed config
    nucleon-seeds.hdf             # seed bank (shared)
  design_0000/
    isobar-conf.yaml              # per-design-point isobar config
    WS1.hdf, WS2.hdf              # nuclear configurations
    trento_events.npy             # cached Trento output
  design_0001/ ...
  results/
    observables.npz               # all observables × all design pts
    summary_table.txt             # human-readable table
    plots/
      corner_lhs.pdf              # LHS space-filling check
      obs_all_panels.pdf          # full observable grid
      obs_vs_R.pdf, ...           # per-parameter scatter plots
```

4.9.2 Key Code Sections

The eccentricity cumulant computation (`stage4_eccentricities`) implements eqs. (4.19) to (4.22) as vectorised NumPy operations. The following listing shows the core moment computation and cumulant evaluation:

```python
def _moments(eps, orders=(2, 4, 6, 8)):
    """Compute <eps^n> for each n in a single pass."""
    return {n: np.mean(eps**n) for n in orders}

def _e2_cumulants(m):
    """Return (e{2}, e{4}, e{6}, e{8}) from moment dict m."""
    m2, m4, m6, m8 = m[2], m[4], m[6], m[8]

    e2 = np.sqrt(m2)
```

```
    inner4 = 2.0 * m2**2 - m4          # e{4}^4 = 2<e^2>^2 - <e^4>
    e4 = inner4**0.25 if inner4 >= 0 else np.nan

    c6 = m6 - 9.0*m4*m2 + 12.0*m2**3  # 6-particle cumulant
    e6 = (-c6/4.0)**(1/6) if -c6/4 >= 0 else np.nan

    c8 = (m8 - 16.0*m6*m2 - 18.0*m4**2
          + 144.0*m4*m2**2 - 144.0*m2**4)
    e8 = (c8/33.0)**(1/8) if c8/33 >= 0 else np.nan

    return e2, e4, e6, e8
```

Listing 4.1: Core eccentricity cumulant computation (from `pbpb_ultracentral_scan.py`).

The hexadecapole sign-change observable and nonlinear coupling (eqs. (4.28) and (4.29)) are computed as:

```
# e4{4}^4 = 2<e4^2>^2 - <e4^4>  (negative in ultracentral: the puzzle)
m4_e4  = np.mean(eps4**4)
e4_4c  = float(2 * np.mean(eps4**2)**2 - m4_e4)

# Sign-change trigger: > 2 => nonlinear channel drives v4{4}^4 < 0
nl_r8  = float(m[8] / m[4]**2) if m[4] > 0 else np.nan
```

Listing 4.2: Hexadecapole cumulant and sign-change trigger.

4.9.3 Configuration Flags

Before running, the user must set:

- `SQRTS`: collision energy (2.76 or 5.02, in TeV);
- `ISOBAR_SAMPLER_DIR`: path to the local clone of `mluzum/Isobar-Sampler`;
- `TRENTO_BIN`: path to the compiled TRENTo binary;
- `N_DESIGN`: number of design points (default 100);
- `N_EVENTS`: Trento events per design point (default 5×10^5).

Each stage can be skipped independently with the `RUN_STAGE_*` boolean flags, enabling re-analysis of previously computed events without re-running TRENTo.

4.10 Summary

This chapter has presented the theoretical and computational foundations of a systematic scan of the ^{208}Pb+^{208}Pb ultracentral flow puzzle. The principal elements are:

1. A deformed Woods-Saxon model of the ^{208}Pb nucleus with five free parameters (R, a, β_3, β_4, w) spanning literature-motivated prior ranges (table 4.3).
2. The isobar-sampler seed-bank method of [25], which ensures that configurations generated at different design points share the same underlying nucleon-position fluctuations, isolating the genuine parameter response from statistical noise.
3. The TRENTo entropy-deposition model [20] with fixed Bayesian-calibrated parameters [21, 22].
4. A 100-point maximin Latin Hypercube design in five dimensions, ensuring uniform marginal coverage and good pairwise separation (algorithm 1).
5. A complete set of eleven eccentricity observables including the four puzzle-critical quantities T_{32}, R_{42}, Γ_2, and $\varepsilon_4\{4\}^4$ (table 4.1).
6. Bootstrap uncertainty estimation with $N_{\text{boot}} = 300$ resamples providing honest $\pm 1\sigma$ bands on all observables (algorithm 2).

The output design matrix, observables, and their uncertainties will serve as training data for a Gaussian Process emulator in the following chapter, enabling a full Bayesian posterior analysis of the puzzle parameters.

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